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United States Patent	12414256
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	An; Chen et al.

Cam latching device

Abstract

A cam latching device is provided for a computing system being installed in a rack. The cam latching device can include a cam and a locking mechanism. The cam includes a handle portion and an engagement portion. The cam includes a locking surface formed thereon. Rotation of the handle portion causes the engagement portion to rotate. The locking mechanism includes a push component and a lock. The push component abuts against a button disposed on a rail of the rack as the computing system translates along a longitudinal axis. When the push component slides against the button, the locking mechanism transitions from a locked configuration to an unlocked configuration. In the locked configuration, the lock abuts against the locking surface to prevent the rotation of the cam. In the unlocked configuration, the lock is retracted away from the locking surface so that the cam is rotatable.

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Appl. No.:	18/446013
Filed:	August 08, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250056752 A1	Feb. 13, 2025

Publication Classification

Int. Cl.: H05K7/14 (20060101); F16B2/18 (20060101)

U.S. Cl.:

CPC **H05K7/1489** (20130101); **F16B2/18** (20130101);

Field of Classification Search

CPC: H05K (7/1409); H05K (7/1489)

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Background/Summary

FIELD

(1) The present disclosure relates generally to a cam latching device operable to assist with installation of a computing system in a rack.

BACKGROUND

(2) As racks become increasingly low-profile, traditional mounting features are no longer

compatible with newer racks that lack front mount EIA posts. To address this issue, heavy HPCs typically use a cam handle to facilitate server engagement in the end position, ensuring stable power and data connections.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Implementations of the present technology will now be described, by way of example only, with reference to the attached figures, wherein:
- (2) FIG. 1A illustrates server system with a rack having a computing system installed therein;
- (3) FIG. 1B illustrates a computing system with a cam latching device;
- (4) FIG. 2A illustrates a cam latching device interacting with a rail of the rack;
- (5) FIG. 2B illustrates the cam latching device;
- (6) FIG. 3A illustrates the computing system with the cam latching device being inserted into the rack;
- (7) FIG. 3B illustrates the cam latching device as utilized in FIG. 3A;
- (8) FIG. 3C illustrates a top view of the cam latching device as in FIGS. 3A and 3B;
- (9) FIG. 4A illustrates the computing system fully inserted into the rack;
- (10) FIG. 4B illustrates the cam latching device as utilized in FIG. 4A;
- (11) FIG. 4C illustrates a top view of the cam latching device as in FIGS. 4A and 4B;
- (12) FIG. 5A illustrates the cam latching device transitioned to a locked configuration;
- (13) FIG. 5B illustrates the cam latching device as utilized in FIG. 5A;
- (14) FIG. 5C illustrates a top view of the cam latching device as in FIGS. 5A and 5B;
- (15) FIGS. 6A and 6B illustrate another examples of a cam latching device;
- (16) FIG. 7A illustrates the cam latching device as the computing system is inserted into the rack;
- (17) FIG. 7B illustrates an opposing view of FIG. 7A;
- (18) FIG. 8A illustrates the cam latching device as the computing system is fully inserted into the rack;
- (19) FIG. 8B illustrates an opposing view of FIG. 8A;
- (20) FIG. 9A illustrates the cam latching device transitioned to the locked configuration; and
- (21) FIG. 9B illustrates an opposing view of FIG. 9A.

DETAILED DESCRIPTION

(22) It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale and the proportions of certain parts may be exaggerated to better illustrate details and features of the present disclosure.

(23) Several definitions that apply throughout this disclosure will now be presented. The term “coupled” is defined as connected, whether directly or indirectly through intervening components, and is not necessarily limited to physical connections. The term “substantially” is defined to be essentially conforming to the particular dimension, shape or other word that substantially modifies, such that the component need not be exact. For example, substantially cylindrical means that the object resembles a cylinder, but can have one or more deviations from a true cylinder. The term “about” means reasonably close to the particular value. For example, about does not require the

exact measurement specified and can be reasonably close. As used herein, the word “about” can include the exact number. The term “near” as used herein is within a short distance from the particular mentioned object. The term “near” can include abutting as well as relatively small distance beyond abutting. The terms “comprising,” “including” and “having” are used interchangeably in this disclosure. The terms “comprising,” “including” and “having” mean to include, but not necessarily be limited to the things so described.

(24) Computing systems such as integrated just a bunch of discs (JBOD) or just a bunch of GPUs (JBOG) need a large insertion and ejection force to allow power bar clip connection to a rack. Conventional latching devices do not provide enough mechanical advantage so that insertion force has a high chance of exceeding 50 pounds. Cam handles can facilitate server engagement in the end position, ensuring stable power and data connections.

(25) The cam latching device disclosed herein includes a long moment arm with a handle portion for enhanced mechanical advantage, allowing for easier system engagement and ensuring electrical connection quality for system integrator improvement and power connection. Additionally, for liquid cooling high performing computing systems, the cam latching device can also ensure easy quick disconnect mating and maintain connecting quality. Conventional cam devices may inadvertently cause the handle portion to rotate towards a closed position during sliding of the computing system in the rack before the cam is fully engaged, potentially leading to conflicts and cam damage.

(26) To resolve this challenge, the present disclosure provides a cam latching device that has a locking mechanism that enables the handle portion to remain open during translation, sliding, or insertion of the computing system into the rack. The handle portion, and consequently the engagement portion, only rotates after reaching the actuating point on the rack when the computing system is fully received in the rack. The present cam latching device ensures a smooth cam engagement process, preventing damage, cable grabbing (when the cam hook travels across the cable management channel), and conflicts.

(27) FIG. 1A illustrates a server system **10** that includes a rack **12** that is operable to receive one or more computing systems **50**. The computing system **50** can include, for example, integrated just a bunch of discs (JBOD) or just a bunch of GPUs (JBOG). The rack **12** can include a plurality of rails **14** that are operable to receive the computing system(s) **50**. The computing systems **50** can be received by the corresponding rails **14**, and translated along the rails **14** along a longitudinal axis until fully received in the rack **12**. Once fully received in the rack **14**, the computing system **50** can then be fully installed by connecting components, for example liquid cooling (e.g., via quick disconnect mating).

(28) As illustrated in FIG. 1B, the computing system **50** can include a chassis **52**. The chassis **52** can have sides **54** and a front **56** that spans the sides **54**. The chassis **52** can be operable to receive a plurality of computing components, such as one or more processors, one or more motherboards, one or more fans, one or more hard drives, etc.

(29) A cam latching device **100** can be coupled with the computing system **50**. In at least one example, the cam latching device **100** can be coupled with the chassis **52**. A handle portion **104** of a cam **102** for the cam latching device **100** can extend from the front **56** of the chassis **52**.

Accordingly, a user can utilize a handle portion **104** of a cam **102** to insert the computing system **50** into the rack **12** and impart a push force to translate the chassis **52** of the computing system **50**, which in some examples can weigh over 300 pounds, along the longitudinal axis into the rack **12** for installation.

(30) FIG. 2A illustrates the cam latching device **100**. As shown in FIG. 2A, the cam latching device **100** includes a cam **102** and a locking mechanism **200**. The cam **102** includes the handle portion **104** and an engagement portion **110**. In some examples, the handle portion **104** can be coupled with the engagement portion **110**. In some examples, the handle portion **104** and the engagement portion **110** can be one integrated component. The handle portion **104** is configured to allow a user to grip

the handle portion **104** and impart the push force, for example exceeding 50 pounds, to translate the computing system **50** along the rails **14** of the rack **12**. The handle portion **104** can have a chassis surface **106** which is operable to abut against the front **56** of the chassis **52** when the handle portion **104** is rotated to couple the computing system **50** with the rack **12** for installation via the engagement portion **110**. In at least one example, a receiving component **60** can be coupled to the front **56** of the chassis **52**. In some examples, the receiving component **60** can be formed as a portion of the front **56** of the chassis **52**. The receiving component **60** can be operable to receive and abut against an abutment surface **108** of the handle portion **104** when the handle portion **104** is rotated to couple the computing system **50** with the rack **12** for installation via the engagement portion **110**.

(31) The engagement portion **110** can be rotatably coupled with the chassis **52**. In at least one example, as illustrated in FIG. 2A, the engagement portion **110** can be rotatably coupled with a side **54** of the chassis **52**. In some examples, the engagement portion **110** can be rotatably coupled with the chassis **52** via a rotation point **112**. In some examples, the rotation point **112** can include any one of the following: a bolt, a screw, a nail, or any other suitable mechanism. When the handle portion **104** is rotated, the engagement portion **110** can be rotated about the rotation point **112**.

(32) The engagement portion **110** can be operable to engage with a guide **80** included on the rack **12**. In at least one example, the guide **80** can be coupled with the rail **14**. When the engagement portion **110** is engaged with the guide **80**, the computing system **50** is latched onto the rack **12** for secure installation.

(33) In at least one example, the engagement portion **110** can form an engagement surface **114** operable to receive and interact with the guide **80**. The engagement surface **114** can have a compound curve so that when the engagement surface **114** of the engagement portion **110** engages with the guide **80** and transitions towards a latched configuration, the cam **102** translates the chassis **52** of the computing system **50** along the longitudinal axis. In other words, the cam **102** converts the rotational force to a linear force due to the compound curve of the engagement surface **114**.

(34) FIG. 2B illustrates the locking mechanism **200**. The locking mechanism **200** includes a push component **204** with a push surface **2040** (as shown in FIGS. 3C, 4C, 5C, 7A, 8A, and 9A) and a lock **210**. The push surface **2040** of the push component **204** can be operable to abut against a button **202** disposed on the rail **14** of the rack **12** as the computing system **50** translates along the longitudinal axis to be installed in the rack **12**. When the push surface **2040** of the push component **204** slides against the button **202**, the locking mechanism **200** transitions from a locked configuration to an unlocked configuration. When the locking mechanism **200** is in the locked configuration, the lock **210** is operable to abut against a locking surface **1100** of the engagement portion **110** of the cam **102** to prevent the rotation of the handle portion **104** and the engagement portion **110**. When the locking mechanism **200** is in the unlocked configuration, the lock **210** is retracted away from the locking surface **1100** so that the handle portion **104** and the engagement portion **110** are rotatable.

(35) In at least one example, as illustrated in FIG. 2B, the locking mechanism **200** can include one or more springs **208**. While FIG. 2B illustrates two springs **208**, the locking mechanism **200** can include one, three, or more springs **208** without deviating from the scope of the disclosure. When the push surface **2040** of the push component **204** slides against the button **202** to transition the locking mechanism **200** to the unlocked configuration, the springs **208** can be compressed. When the push component **204** does not contact the button **202**, the springs **208** can be not compressed and maintain the position of the lock **210** against the locking surface **1100** of the cam **102**. As the lock **210** moves in coordination with the springs **208**, the lock **210** retracts when the springs **208** are compressed and extends when the springs **208** are not compressed. In some examples, the lock **210** can be coupled with the push component **204**. In some examples, as illustrated in FIG. 2B, the locking mechanism **200** can include a body **206** coupled with the push component **204** and the lock

210. The body **206** can be configured to receive the springs **208**. Accordingly, when the push component **204** slides against the button **202** to transition the locking mechanism **200** to the unlocked configuration, the body **206** pushes against the one or more springs **208** to compress the springs **208** and retracts the lock **210** from the locking surface **1100**. In some examples, when the lock **210** is retracted, the lock **210** can be moved towards the chassis **52** of the computing system **50**. In some examples, the lock **210** can change in shape and/or size, to be retracted away from the locking surface **1100**. For example, the lock **210** can telescope to be extended out to abut against the locking surface **1100** and retract back to a smaller length to not abut against the locking surface **1100**.

(36) In at least one example, the locking mechanism **200** can be coupled with the computing system **50**, for example the chassis **52**. In some examples, the locking mechanism **200** can be coupled with the side **54** of the chassis **52**. In some examples, the locking mechanism **200** can be coupled with the cam **102**. For example, as illustrated in FIG. 2B, the locking mechanism **200** can be coupled with the engagement portion **110** of the cam **102**.

(37) FIGS. 3A, 3B, and 3C illustrate the cam latch device **100** being utilized by a user to push the computing system **50** into the rack **12** for installation. The user is imparting the push force on the handle portion **104** of the cam **102** to translate the computing system **50** along the longitudinal axis into the rack **12**. The locking mechanism **200** is in the locked configuration so that the handle portion **104** cannot rotate. Accordingly, the user can effectively impart the push force onto the computing system **50** without damaging the cam **102**, the rail **14**, the rack **12**, and/or any other components. The push force can then be translated from the user to the handle portion **104** to the chassis **52** of the computing system **50** effectively, efficiently, and consistently.

(38) As the chassis **52** of the computing system **50** has not yet been fully received in the rack **12**, the push surface **2040** of the push component **204** has not yet contacted the button **202** on the rack **12**. Accordingly, the locking component **200** is in the locked configuration such that the handle portion **104** and the engagement portion **110** are prevented from rotating. The lock **210** abuts against the locking surface **1100** of the cam **102** to prevent the rotation of the cam **102**. For example, as illustrated in FIG. 3C, the springs **208** are not compressed, so the lock **210** is extended and abuts against the locking surface **1100**. The locking mechanism **200**, with the springs **208**, defaults to the locked configuration to prevent rotation of the handle portion **104** and the engagement portion **110** which can avoid the engagement portion **110** from getting caught by the guide **80**.

(39) FIGS. 4A, 4B, and 4C illustrate the chassis **52** of the computing system **50** being almost fully received in the rack **12** for installation. The button **202** is abutting against and sliding across the push surface **2040** of the push component **204**. In at least one example, as illustrated in FIG. 4C, the push surface **2040** can be angled. In some examples, the push surface **2040** can be curved. Accordingly, the button **202** can abut against and smoothly slide across the push surface **2040** of the push component **204** to compress the springs **208**. As the springs **208** are compressed, the push component **204** and the body **206** retract towards the chassis **52** (e.g., away from the rail **14**). As the lock **210** is corresponding with the push component **204** and/or the body **206**, the lock **210** begins to retract away from the locking surface **1100** to transition the locking mechanism **200** towards the unlocked configuration. As illustrated in FIG. 4C, the lock **210**, similar to the push component **204** and/or the body **206**, moves towards the chassis **52** and away from the rail **14**. Once the lock **210** is fully separated from the locking surface **1100**, the locking mechanism **200** is in the unlocked configuration, and the cam **102** (e.g., the handle portion **104** and the engagement portion **110**) can rotate to latch the computing system **50** in the rack **14**.

(40) FIGS. 5A, 5B, and 5C illustrate the chassis **52** of the computing system **50** being fully inserted into the rack **14**, and the cam **102** has latched the computing system **50** in the rack **14**. Accordingly, the computing system **50** is coupled with the rack **14**. As shown in FIG. 5C, the button **202** is fully abutting the push component **204**, the springs **208** are compressed, and the lock **210** is separated

and not abutting the locking surface **1100**.

(41) FIGS. **6A-9B** illustrate another example of a cam latching device **100**. The cam latching device **100** in FIGS. **6A-9B** have the same components and operation as the cam latching device **100** as shown and described in FIGS. **3A-5C**, but is oriented so that the handle portion **104** is horizontal instead of vertical as in FIGS. **3A-5C**. This slightly alters the arrangement of the components of the cam latching device **100**, but does not alter the operation of the components.

(42) As illustrated in FIGS. **6A** and **6B**, the cam latching device **100** includes the cam **102** which includes the handle portion **104** and the engagement portion **110**.

(43) FIGS. **7A** and **7B** are similar to FIGS. **3A-3C**, the cam latch device **100** is being utilized by a user to push the computing system **50** into the rack **12** for installation, as the computing system **50** has not yet been fully received in the rack **12**. The locking mechanism **200** is in the locked configuration so that the handle portion **104** cannot rotate. Accordingly, the user can effectively impart the push force onto the computing system **50** without damaging the cam **102**, the rail **14**, the rack **12**, and/or any other components. The push force can then be translated from the user to the handle portion **104** to the chassis **52** of the computing system **50** effectively, efficiently, and consistently.

(44) As the chassis **52** of the computing system **50** has not yet been fully received in the rack **12**, the push surface **2040** of the push component **204** has not yet contacted the button **202** on the rack **12**. Accordingly, the locking component **200** is in the locked configuration such that the handle portion **104** and the engagement portion **110** are prevented from rotating. The lock **210** abuts against the locking surface **1100** of the cam **102** to prevent the rotation of the cam **102**. For example, as illustrated in FIGS. **7A** and **7B**, the springs **208** are not compressed, so the lock **210** is extended and abuts against the locking surface **1100**. The locking mechanism **200**, with the springs **208**, defaults to the locked configuration to prevent rotation of the handle portion **104** and the engagement portion **110** which can avoid the engagement portion **110** from getting caught by the guide **80**.

(45) FIGS. **8A** and **8B** illustrate the chassis **52** of the computing system **50** being almost fully received in the rack **12** for installation. The button **202** is abutting against and sliding across the push surface **2040** of the push component **204**. In at least one example, as illustrated in FIGS. **8A**, the push surface **2040** can be angled. In some examples, the push surface **2040** can be curved. Accordingly, the button **202** can abut against and smoothly slide across the push surface **2040** of the push component **204** to compress the springs **208**. As the springs **208** are compressed, the push component **204** and the body **206** retract towards the chassis **52** (e.g., away from the rail **14**). As the lock **210** is corresponding with the push component **204** and/or the body **206**, the lock **210** begins to retract away from the locking surface **1100** to transition the locking mechanism **200** towards the unlocked configuration. As illustrated in FIG. **8B**, the lock **210**, similar to the push component **204** and/or the body **206**, moves towards the chassis **52** and away from the rail **14**. Once the lock **210** is fully separated from the locking surface **1100**, the locking mechanism **200** is in the unlocked configuration, and the cam **102** (e.g., the handle portion **104** and the engagement portion **110**) can rotate to latch the computing system **50** in the rack **14**.

(46) FIGS. **9A** and **9B** illustrate the chassis **52** of the computing system **50** being fully inserted into the rack **14**, and the cam **102** has latched the computing system **50** in the rack **14**. Accordingly, the computing system **50** is coupled with the rack **14**. As shown in FIG. **9A**, the button **202** is fully abutting the push component **204**, the springs **208** are compressed, and as shown in FIG. **9B**, the lock **210** is separated and not abutting the locking surface **1100**. Accordingly, the cam **102** (e.g., the handle portion **104** and the engagement portion **110**) have been rotated to latch the computing system **50** in the rack **14**.

(47) The embodiments shown and described above are only examples. Even though numerous characteristics and advantages of the present technology have been set forth in the foregoing description, together with details of the structure and function of the present disclosure, the

disclosure is illustrative only, and changes may be made in the detail, especially in matters of shape, size and arrangement of the parts within the principles of the present disclosure to the full extent indicated by the broad general meaning of the terms used in the attached claims. It will therefore be appreciated that the embodiments described above may be modified within the scope of the appended claims.

Claims

1. A cam latching device for a computing system being installed in a rack, the cam latching device comprising: a cam including a handle portion and an engagement portion, the cam including a locking surface formed thereon, wherein rotation of the handle portion causes the engagement portion to rotate; and a locking mechanism including a push component and a lock, the push component operable to abut against a button disposed on a rail of the rack as the computing system translates along a longitudinal axis to be installed in the rack, wherein when the push component slides against the button, the locking mechanism transitions from a locked configuration to an unlocked configuration, wherein when the locking mechanism is in the locked configuration, the lock is operable to abut against the locking surface to prevent the rotation of the handle portion and the engagement portion, and wherein when the locking mechanism is in the unlocked configuration, the lock is retracted away from the locking surface so that the handle portion and the engagement portion are rotatable.
2. The cam latching device of claim 1, wherein the locking mechanism includes one or more springs, wherein when the push component slides against the button to transition the locking mechanism to the unlocked configuration, the springs are compressed.
3. The cam latching device of claim 2, wherein when the push component does not contact the button, the one or more springs are not compressed and maintain the position of the lock against the locking surface of the cam.
4. The cam latching device of claim 2, wherein the locking mechanism includes a body coupled with the push component and the lock, wherein the body is configured to receive the one or more springs.
5. The cam latching device of claim 4, wherein when the push component slides against the button to transition the locking mechanism to the unlocked configuration, the body pushes against the one or more springs to compress the one or more springs and retracts the lock from the locking surface.
6. The cam latching device of claim 1, wherein when the handle portion is rotated, the engagement portion is rotated to engage with a guide to maintain the position of the computing system in the rack.
7. The cam latching device of claim 1, wherein the locking mechanism is only transitioned to the unlocked configuration when the computing system is fully received into the rack.
8. A computing system operable to be installed in a rack, the computing system comprising: a chassis; and a cam latching device coupled with the chassis, the cam latching device including: a cam including a handle portion and an engagement portion, the cam including a locking surface formed thereon, wherein rotation of the handle portion causes the engagement portion to rotate; and a locking mechanism including a push component and a lock, the push component operable to abut against a button disposed on a rail of the rack as the computing system translates along a longitudinal axis to be installed in the rack, wherein when the push component slides against the button, the locking mechanism transitions from a locked configuration to an unlocked configuration, wherein when the locking mechanism is in the locked configuration, the lock is operable to abut against the locking surface to prevent the rotation of the handle portion and the engagement portion, and wherein when the locking mechanism is in the unlocked configuration, the lock is retracted away from the locking surface so that the handle portion and the engagement portion are rotatable.

9. The computing system of claim 8, wherein the locking mechanism includes one or more springs, wherein when the push component slides against the button to transition the locking mechanism to the unlocked configuration, the springs are compressed.
 10. The computing system of claim 9, wherein when the push component does not contact the button, the one or more springs are not compressed and maintain the position of the lock against the locking surface of the cam.
 11. The computing system of claim 9, wherein the locking mechanism includes a body coupled with the push component and the lock, wherein the body is configured to receive the one or more springs.
 12. The computing system of claim 11, wherein when the push component slides against the button to transition the locking mechanism to the unlocked configuration, the body pushes against the one or more springs to compress the one or more springs and retracts the lock from the locking surface.
 13. The computing system of claim 8, wherein when the handle portion is rotated, the engagement portion is rotated to engage with a guide disposed on the rack to maintain the position of the computing system in the rack.
 14. The computing system of claim 13, wherein when the engagement portion is engaged with the guide, the handle portion abuts against the chassis.
 15. The computing system of claim 8, wherein the locking mechanism is only transitioned to the unlocked configuration when the chassis is fully received into the rack.
 16. The computing system of claim 8, wherein the handle portion is operable to receive a push force from a user to translate the chassis along the longitudinal axis into the rack.
 17. A server system comprising: a rack including a plurality of rails and a button disposed on at least one of the rails; a computing system operable to be installed into the rack, the computing system including: a chassis operable to be received by the rails; and a cam latching device coupled with the chassis, the cam latching device including: a cam including a handle portion and an engagement portion, the cam including a locking surface formed thereon, wherein rotation of the handle portion causes the engagement portion to rotate; and a locking mechanism including a push component and a lock, the push component operable to abut against the button as the computing system translates along a longitudinal axis to be installed in the rack, wherein when the push component slides against the button, the locking mechanism transitions from a locked configuration to an unlocked configuration, wherein when the locking mechanism is in the locked configuration, the lock is operable to abut against the locking surface to prevent the rotation of the handle portion and the engagement portion, and wherein when the locking mechanism is in the unlocked configuration, the lock is retracted away from the locking surface so that the handle portion and the engagement portion are rotatable.
 18. The server system of claim 17, wherein the locking mechanism includes one or more springs, wherein when the push component slides against the button to transition the locking mechanism to the unlocked configuration, the springs are compressed.
 19. The server system of claim 18, wherein when the push component does not contact the button, the one or more springs are not compressed and maintain the position of the lock against the locking surface of the cam.
 20. The server system of claim 17, wherein when the handle portion is rotated, the engagement portion is rotated to engage with a guide disposed on the rack to maintain the position of the computing system in the rack.
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